

2021-Ph-H1-Q10

Section: Our Dynamic Universe

Topic: Relativity

A spaceship moves at $0.8c$.

Which statement is true?

Solution:

At high speed:

- Time **dilates**
- Length **contracts**
- Mass **increases**

Answer: D

Guidance for Students:

When moving at relativistic speeds:

- Time runs **slower** (to an outside observer)
- Length **contracts**
- Mass appears **greater**

Revision Tip:

Learn this trio:

Time , Length , Mass as $v \rightarrow c$